

A mathematical model

A mathematical model represents a situation using quantities and their relationships.

What assumptions should we identify before writing an equation?

Choose a variable

A printing service charges \$12 plus \$0.08 per page.

Which variable and units would describe an order's total cost?

A printing budget

A printing service charges \$12 plus \$0.08 per page. Your budget is exactly \$28.

How many pages can you order?

Translate a relationship

One number is 7 more than twice another.
Their sum is 43.

What equation finds the smaller number?

An average target

Four equally weighted quiz scores are 76, 84, 91, and 85. One quiz remains.

What fifth score would produce an average of 86?

Compare two plans

Plan A costs \$18 plus \$4 per visit. Plan B costs \$30 plus \$2 per visit.

At how many visits do the plans cost the same?

A percent change

A book's sale price is \$36 after a 20% discount. Tax is excluded.

What was its original price?

A mixture

Mix a 10% salt solution with a 30% salt solution to make 8 liters of a 25% solution. Assume volumes add.

How many liters of the 10% solution are needed?

Distance, rate, and time

A bus travels at a constant 48 miles per hour for 2.5 hours.

How far does it travel?

Two travel rates

A 150-mile trip takes 3 hours: part at 40 mph and part at 60 mph. Ignore stops.

How long is spent traveling at 40 mph?

A perimeter model

A rectangle's length is 5 meters more than its width. Its perimeter is 42 meters.

What is the width?

Use a geometry formula

A triangle has area 54 square centimeters and base 12 centimeters.

What is its height?

A work-rate model

One machine completes a job in 6 hours; another completes it in 3 hours. Their constant rates add.

How long would they take working together?

Check the model

A ticket order costs \$11 per ticket plus a \$7 fee. The total is \$52.

Does the model give a feasible ticket count?

Interpret a restriction

A rental costs \$25 plus \$3 per hour, with partial hours charged proportionally. The rental lasts at most 8 hours.

What input values and total costs are possible?